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Total No. of Questions : 13

Total No. of Pages :04

B.Pharmacy. (Sem.-7)

NOVEL DRUG DELIVERY SYSTEM-THEORY

Subject Code :BP-704T

M.Code : 78390

Date of Examination :28-11-2025

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TWENTY questions carrying ONE marks each.
2. SECTION-B contains THREE questions carryingTEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. **Write briefly :**

a. Which of the following is an advantage of controlled release systems?

i) Reduced patient compliance ii) Reduced dosing frequency

iii) Increased dose dumping iv) Increased drug fluctuation

b. The diffusion mechanism in controlled release systems generally follows:

i) Fick's law ii) Boyle's law

iii) Raoult's law iv) Henry's law

c. Which polymer is commonly used in osmotic pump systems?

i) Cellulose acetate ii) Polyethylene glycol

iii) Gelatin iv) Starch

d. Bioadhesion in mucosal drug delivery is due to:

i) Hydrophobic interactions ii) Electrostatic and hydrogen bonding

iii) Only van der Waals forces iv) Crystallinity

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e. Which of the following is NOT a method of microencapsulation?

i) Coacervation-phase separation
ii) Spray drying
iii) Solvent evaporation
iv) Granulation by slugging

f. What is the main barrier to drug permeation through skin?

i) Dermis
ii) Epidermis
iii) Stratum corneum
iv) Sebaceous gland

g. Floating drug delivery systems rely on:

i) Mucoadhesion
ii) Low density
iii) Osmotic pressure
iv) Efflux transporters

h. Dry powder inhalers are formulated mainly for: .

i) Nasal cavity
ii) Pulmonary delivery
iii) Buccal absorption
iv) Ocular absorption

i. Which of the following is a limitation of targeted drug delivery systems?

i) Higher patient compliance
ii) Reduced toxicity
iii) Complexity of formulation
iv) Site-specific action

j. Liposomes are primarily made of:

i) Proteins
ii) Polysaccharides
iii) Phospholipids
iv) Glycoproteins

k. Which is NOT an ocular barrier?

i) Blood-aqueous barrier
ii) Corneal epithelium
iii) Retinal pigment epithelium
iv) Blood-brain barrier

l. An intrauterine device (IUD) is used mainly for:

i) Local ocular therapy
ii) Contraception
iii) Diabetes management
iv) Pain relief

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m. Which release mechanism involves swelling followed by drug diffusion?

i) Osmotic release
ii) Diffusion
iii) Dissolution-controlled release
iv) Matrix swelling

n. Microspheres differ from microcapsules by:

i) Size
ii) Nature of drug dispersion
iii) Manufacturing method
iv) Use of polymers

o. The main permeation enhancers in TDDS act by:

i) Disrupting lipid bilayers
ii) Increasing pH
iii) Increasing viscosity
iv) Decreasing solubility

p. Osmotic pumps maintain drug release by:

i) Enzymatic degradation
ii) Osmotic pressure gradient
iii) pH modulation
iv) Ion exchange

q. Mucoadhesive polymers include:

i) Chitosan
ii) Starch
iii) Sodium chloride
iv) Silica

r. Nebulizers generate:

i) Dry particles
ii) Aerosolized mist
iii) Crystals
iv) Lipid vesicles

s. Which is a biodegradable polymer?

i) Polyethylene
ii) Polylactic acid
iii) Polyvinyl chloride
iv) Polystyrene

t. A key application of monoclonal antibodies in drug delivery is:

i) Passive diffusion enhancers
ii) Targeted drug delivery
iii) Osmotic pump coating
iv) Solubility enhancers

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SECTION-B

2. Discuss the advantages, disadvantages and selection criteria of drugs for controlled release formulations.

3. Enumerate different methods of microencapsulation and explain in detail coacervation phase separation technique.

4. Write detailed notes on the formulation approaches and components of TDDS.

SECTION-C

5. Differentiate between diffusion-controlled and dissolution-controlled drug delivery systems.

6. List the properties of polymers suitable for controlled release.

7. Write anote on buccal drug delivery systems.

8. Discuss the concept of osmotic pump implants.

9. Write short notes on floating and high-density GRDDS.

10. Explain the formulation of dry powder inhalers.

11. Write a note on niosomes as drug carriers.

12. Discuss ocular barriers relevant to drug delivery.

13. Give applications of intrauterine devices in controlled release.

NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.